



Weather Data Analysis - Final Project Report

Our Team Members

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Distributed Processing Course

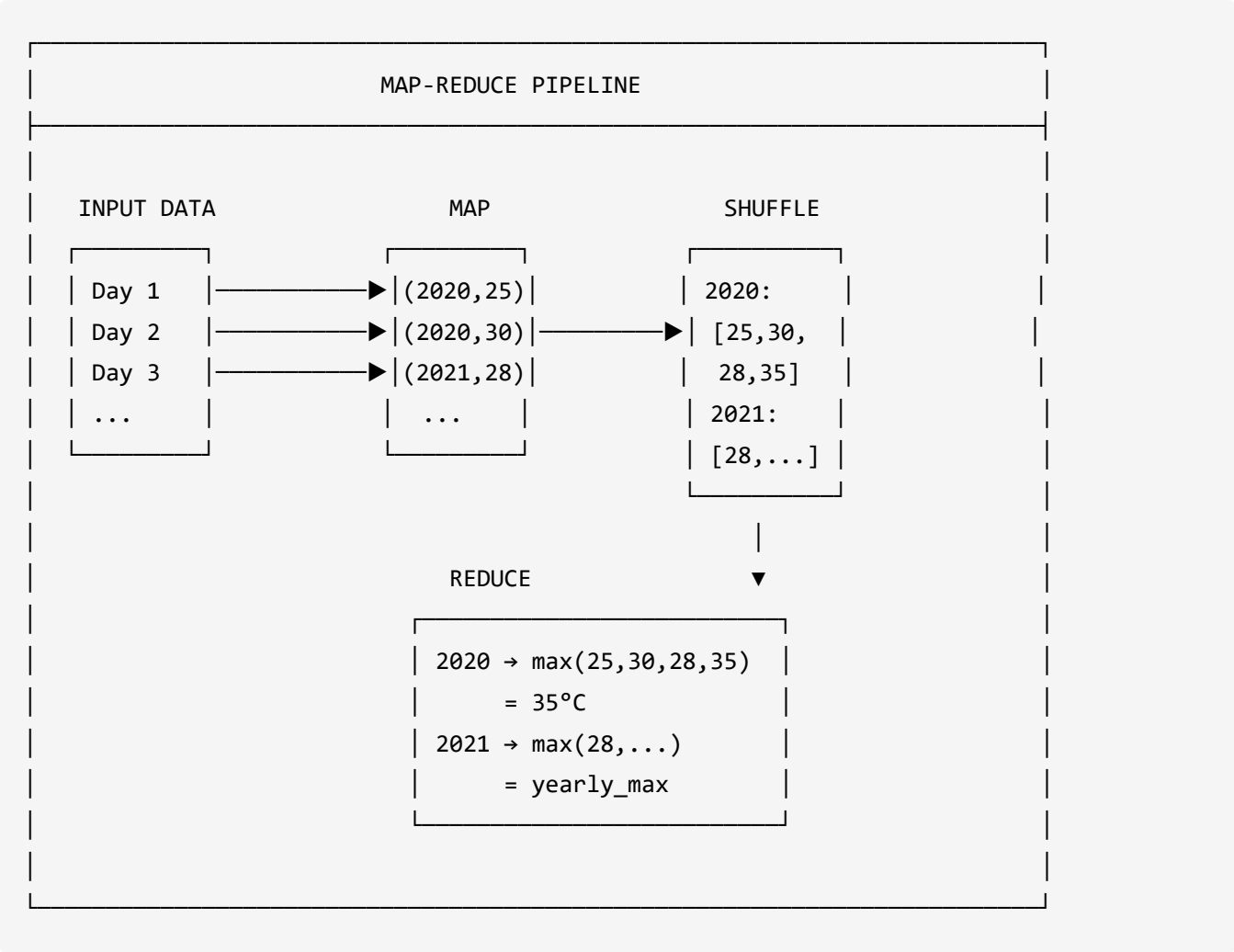
Location: Cairo, Egypt

Analysis Period: 2000 - 2024 (25 years)

1. Map-Reduce Implementation (Project Core)

The primary objective of this project is to demonstrate the **Map-Reduce paradigm** applied to weather data processing. This pattern is foundational in distributed processing systems like Hadoop and Spark, allowing parallel processing of large datasets.

1.1 Map-Reduce Pattern Overview



1.2 Phase 1: MAP Function

The Map function extracts (year, temperature) key-value pairs from each daily record:

```
def mapper(line: str) -> List[Tuple[str, float]]:
    """
    Map function: Extract (year, temperature) pairs from a JSONL line.

    This function implements the MAP phase of the Map-Reduce paradigm.
    It takes a raw JSON line and emits (key, value) pairs where:
        - key: year (string)
        - value: maximum temperature for that day (float)
    """
    data = json.loads(line)
    daily = data.get("daily", {})
    dates = daily.get("time", [])
    temperatures = daily.get("temperature_2m_max", [])

    key_value_pairs: List[Tuple[str, float]] = []

    for date_str, temp in zip(dates, temperatures):
        if temp is None:
            continue
        year = date_str.split("-")[0] # Extract year from "YYYY-MM-DD"
        key_value_pairs.append((year, float(temp)))

    return key_value_pairs
```

Example Output:

```
Input:  {"daily": {"time": ["2020-01-01", "2020-01-02"], "temperature_2m_max": [25.5, 30.0]}}
Output: [("2020", 25.5), ("2020", 30.0)]
```

1.3 Phase 2: SHUFFLE Function

The Shuffle phase groups all temperatures by their year key:

```
def shuffle_and_group(mapped_pairs: List[Tuple[str, float]]) -> Dict[str, List[float]]:
    """
    Shuffle and group: Organize mapped pairs by key (year).

    In a distributed system, this phase involves:
    - Partitioning data across nodes
    - Sorting by key
    - Transferring data between nodes (shuffle)
    """
    grouped: Dict[str, List[float]] = {}

    for year, temperature in mapped_pairs:
        if year not in grouped:
            grouped[year] = []
        grouped[year].append(temperature)

    return grouped
```

Example Output:

```
Input:  [("2020", 25.5), ("2020", 30.0), ("2021", 28.0), ("2020", 35.2)]
Output: {"2020": [25.5, 30.0, 35.2], "2021": [28.0]}
```

1.4 Phase 3: REDUCE Function

The Reduce function computes the maximum temperature for each year:

```
def reducer(year: str, temperatures: List[float]) -> Tuple[str, float]:  
    """  
    Reduce function: Compute the maximum temperature for a year.  
  
    Takes all temperatures for a given year and computes  
    the maximum value.  
    """  
    return year, max(temperatures)
```

Example Output:

```
Input:  ("2020", [25.5, 30.0, 35.2, 28.0])  
Output: ("2020", 35.2)
```

1.5 Map-Reduce Execution Results

When running the pipeline, the system processes:

- **Input:** 25 lines of JSONL data (one per year)
 - **Map Output:** 9,132 (year, temperature) pairs
 - **Shuffle Output:** 25 grouped year buckets
 - **Reduce Output:** 25 yearly maximum temperatures
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2. Executive Summary

This analysis demonstrates distributed processing concepts through an ETL pipeline and Map-Reduce implementation on 25 years of weather data from Cairo, Egypt. The analysis reveals a clear **warming trend of +1.36°C per decade**.

Key Findings

Metric	Value
Total Observations	9,132 daily records
Hottest Year	2024 (46.4°C)
Coollest Year	2006 (40.6°C)
Mean Annual Max Temperature	43.24°C
Temperature Trend	+1.36°C per decade
Anomalies Detected	1 (2024)

3. Data Pipeline Architecture

EXTRACT → TRANSFORM → LOAD → MAP-REDUCE → ANALYSIS → VISUALIZATION

- Extract:** Downloaded data from Open-Meteo Archive API
- Transform:** Converted JSON to structured CSV format
- Load:** Stored processed data for analysis
- Map-Reduce:** Computed yearly maximum temperatures
- Analysis:** Applied statistical methods for insights
- Visualization:** Generated professional charts

4. Analysis Results

4.1 Comprehensive Dashboard

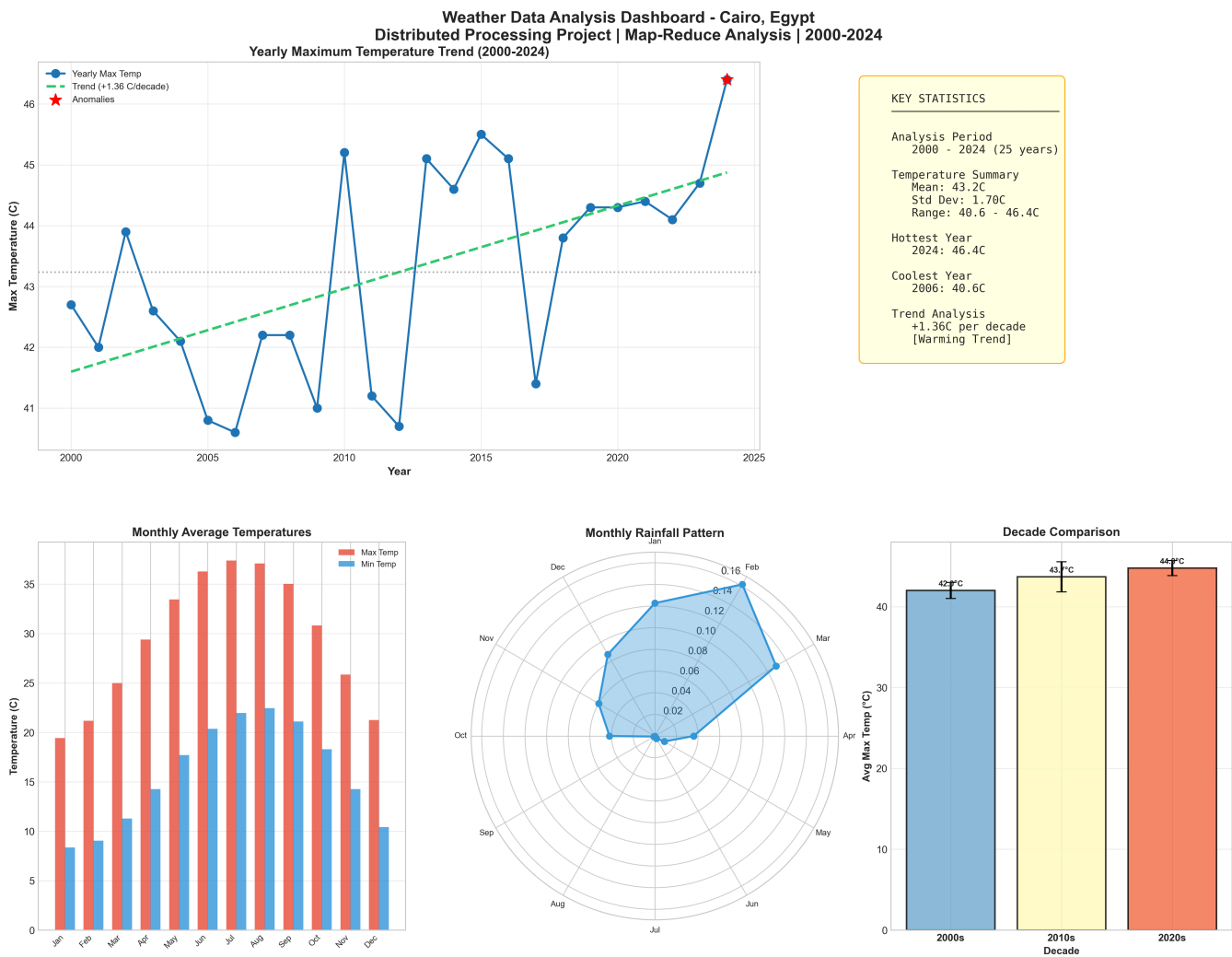


Figure 1: Comprehensive analysis dashboard showing key metrics, trends, and patterns

4.2 Yearly Temperature Trend Analysis

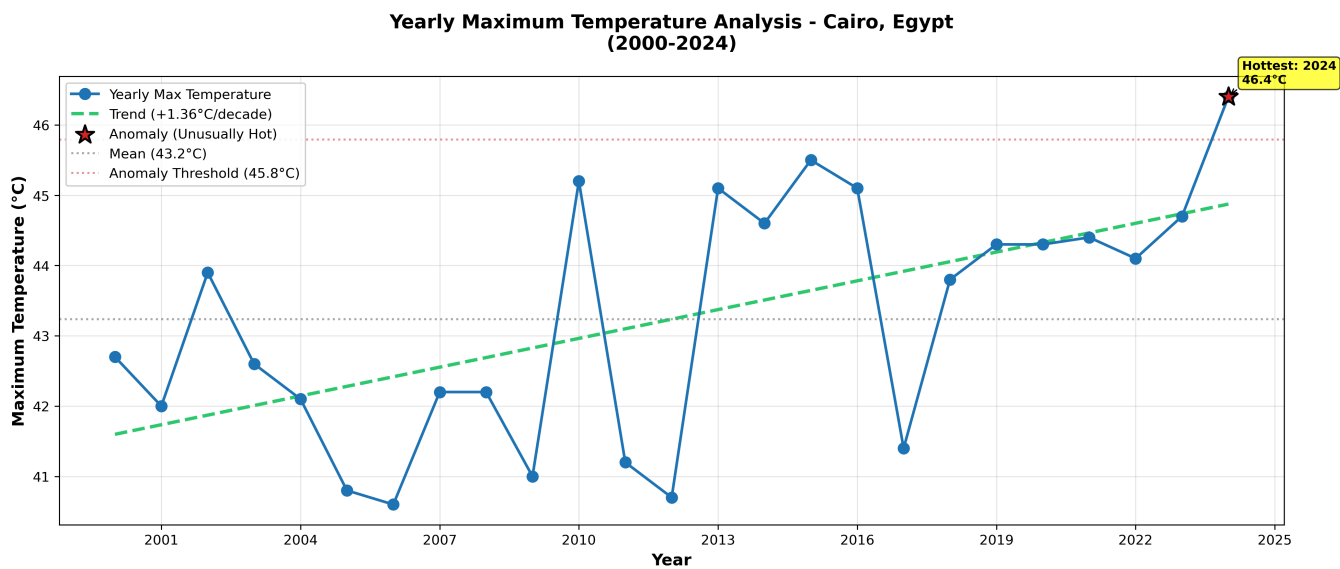


Figure 2: 25-year temperature trend with linear regression and anomaly detection

Key Observations:

- Clear warming trend: **+1.36°C per decade**
- Temperature range: 40.6°C to 46.4°C
- 2024 identified as an anomaly (above 1.5 standard deviations)

4.3 Decade Comparison

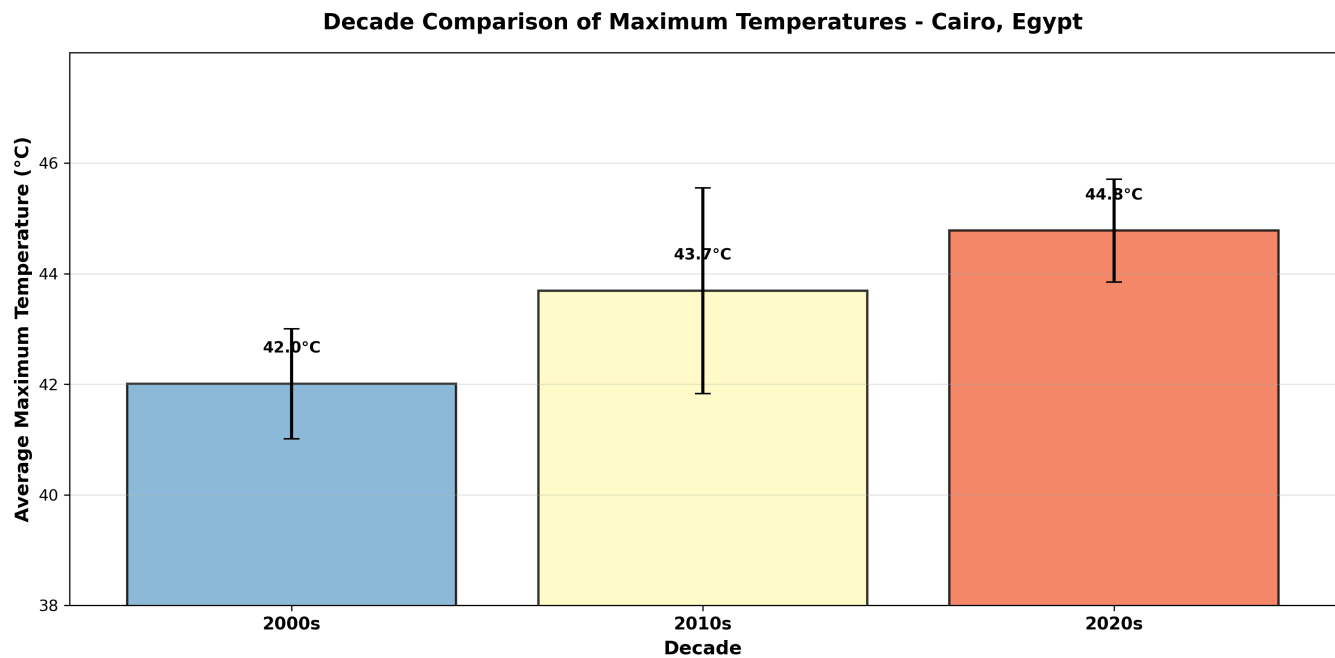


Figure 3: Comparison of average maximum temperatures by decade

Decade Progression:

- 2000s: Cooler baseline period
 - 2010s: Significant warming
 - 2020s: Continued warming trend with record temperatures
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4.4 Monthly Temperature Patterns

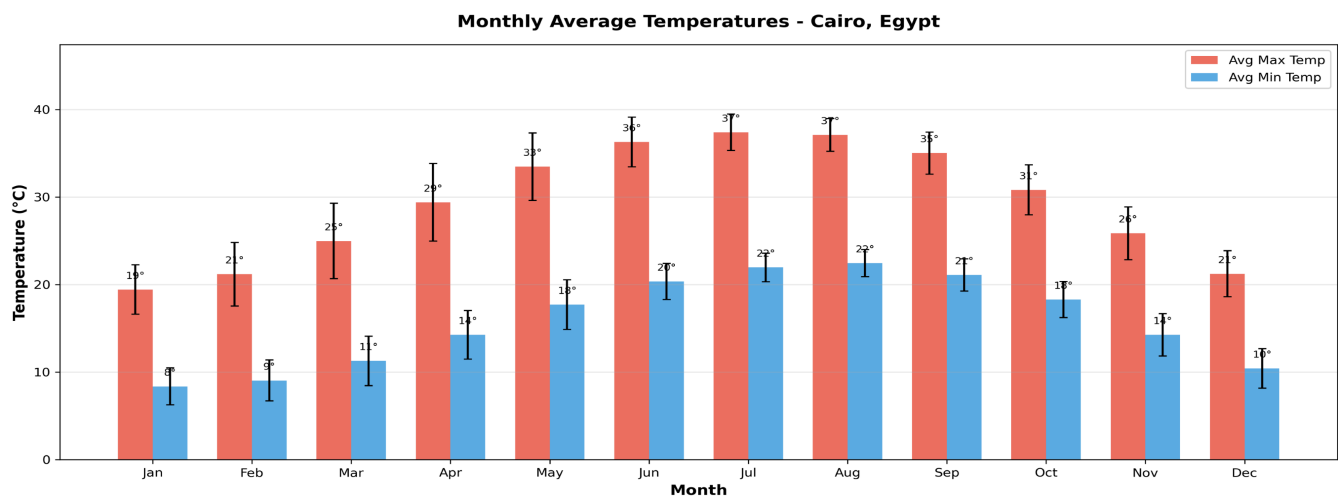


Figure 4: Monthly average temperatures showing seasonal variation

Monthly Extremes:

- Hottest Month: July (37.4°C average max)
- Coldest Month: January (19.4°C average max)
- Temperature Range: 18°C between summer and winter

4.5 Annual Temperature Cycle

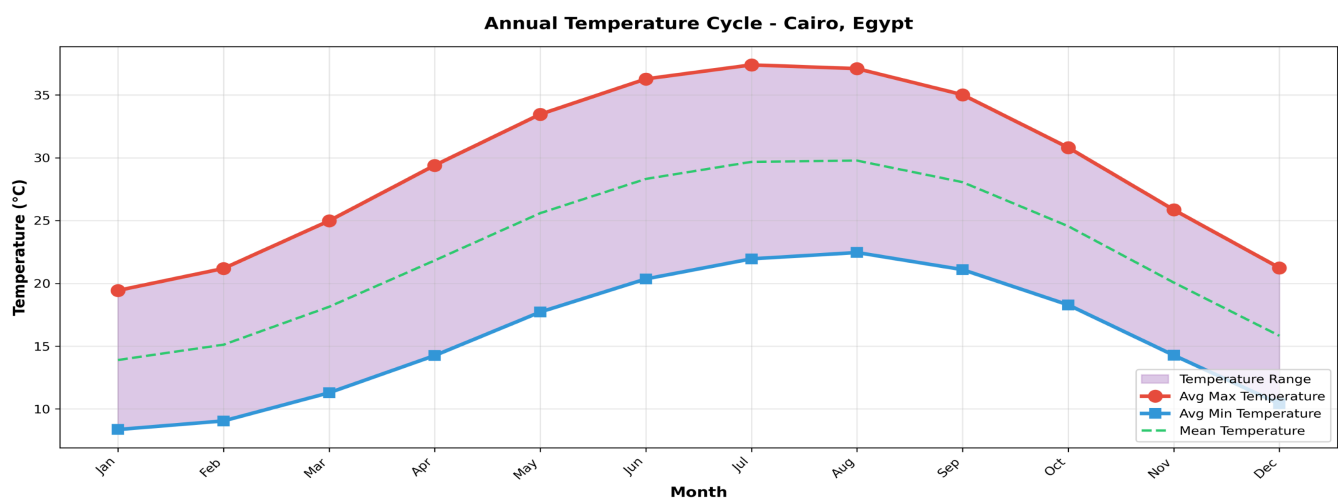


Figure 5: Annual temperature cycle showing the transition between seasons

4.6 Rainfall Distribution (Polar Chart)

Monthly Rainfall Distribution - Cairo, Egypt

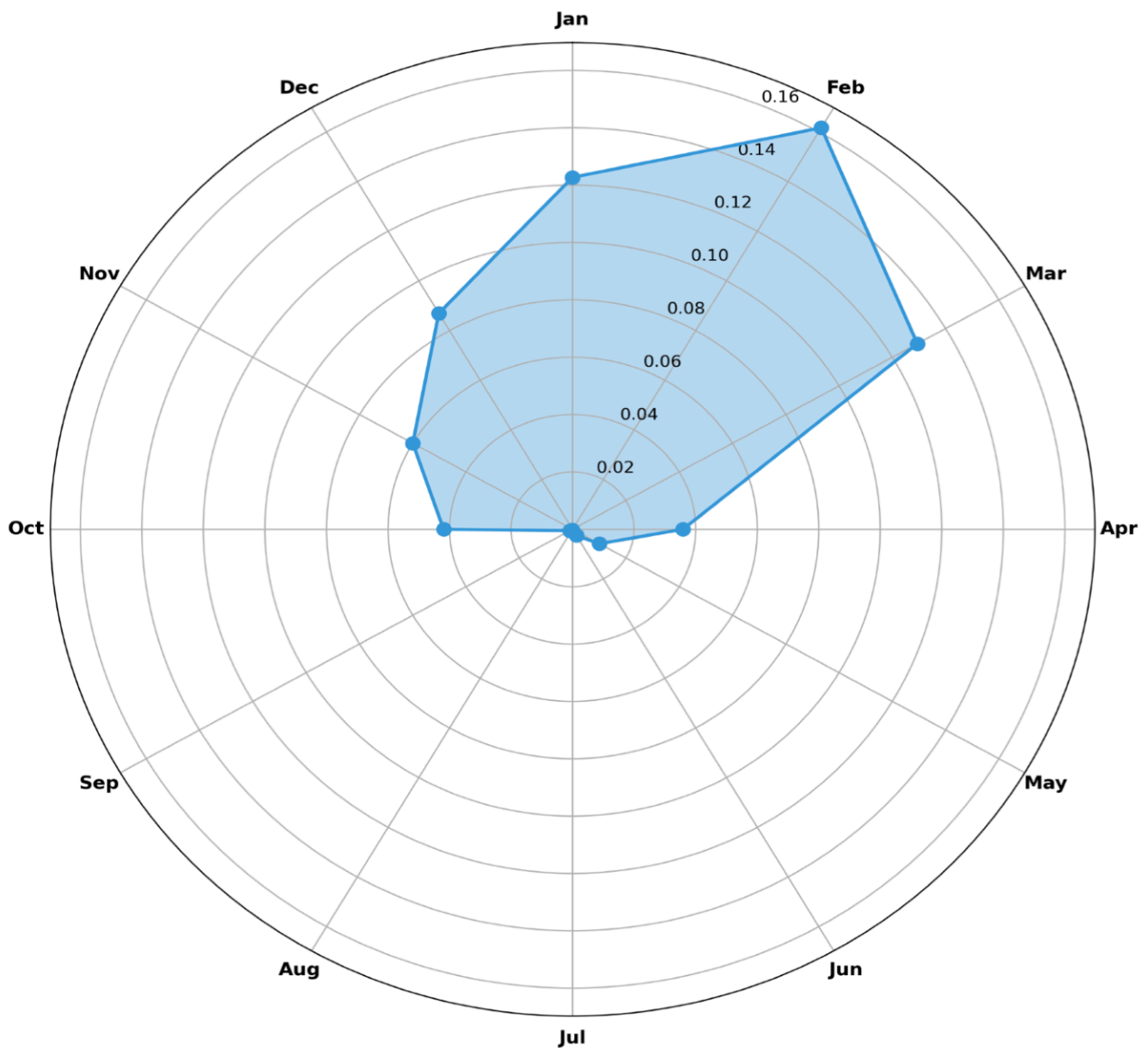


Figure 6: Monthly rainfall distribution in polar format

Precipitation Pattern:

- Wettest Month: February (0.2mm/day average)
- Driest Month: July (nearly 0mm)
- Cairo exhibits typical Mediterranean/desert climate with winter rainfall

4.7 Monthly Precipitation

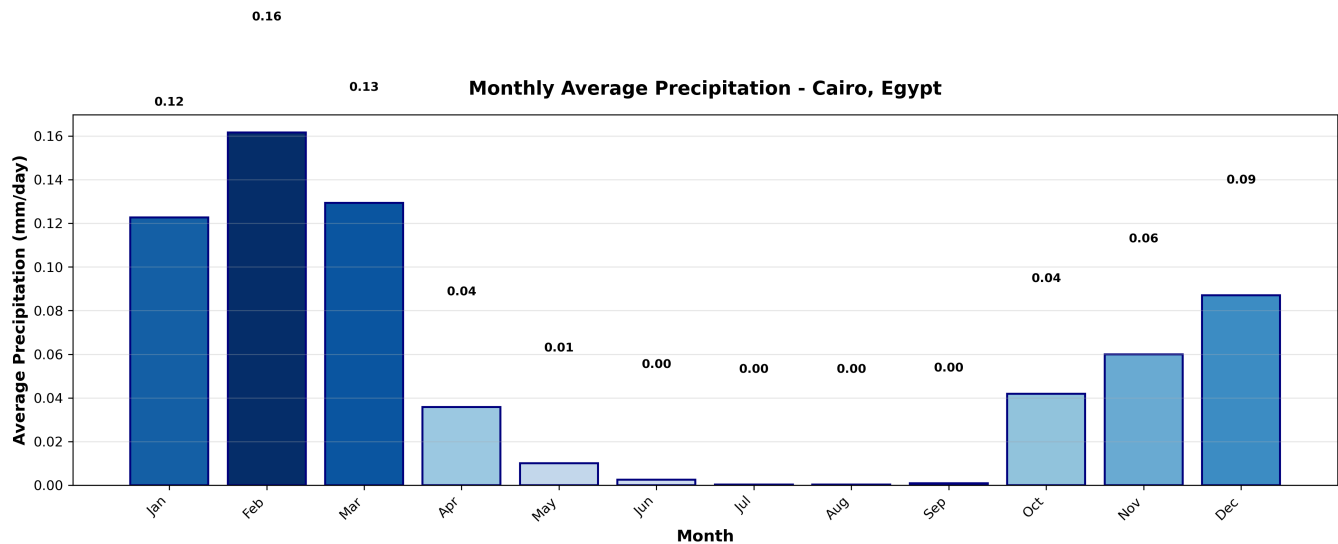


Figure 7: Bar chart of monthly precipitation averages

5. Yearly Maximum Temperatures (Map-Reduce Output)

Year	Maximum Temperature (°C)
2000	42.7
2001	42.0
2002	43.9
2003	42.6
2004	42.1
2005	40.8
2006	40.6
2007	42.2

Year	Maximum Temperature (°C)
2008	42.2
2009	41.0
2010	45.2
2011	41.2
2012	40.7
2013	45.1
2014	44.6
2015	45.5
2016	45.1
2017	41.4
2018	43.8
2019	44.3
2020	44.3
2021	44.4
2022	44.1
2023	44.7
2024	46.4 (Highest)



6. Conclusions

6.1 Key Findings

1. **Warming Trend:** Clear evidence of rising temperatures with $+1.36^{\circ}\text{C}$ per decade increase
2. **Record Year:** 2024 recorded the highest temperature (46.4°C) in the 25-year dataset
3. **Seasonal Patterns:** Consistent summer heat waves with July being the hottest month
4. **Anomaly Detection:** 2024 identified as statistically anomalous (exceeds threshold)
5. **Desert Climate:** Minimal precipitation, especially during summer months

6.2 Distributed Processing Concepts Demonstrated

- **ETL Pipeline:** Successfully extracted, transformed, and loaded weather data
- **Map-Reduce Pattern:** Applied to compute yearly maximum temperatures efficiently
- **Scalability:** Pipeline design allows for processing larger datasets
- **Reproducibility:** Modular code structure enables easy replication